

Demonstrating Passive yet Interactive Soft Haptic Patches Anywhere using MagnetIO

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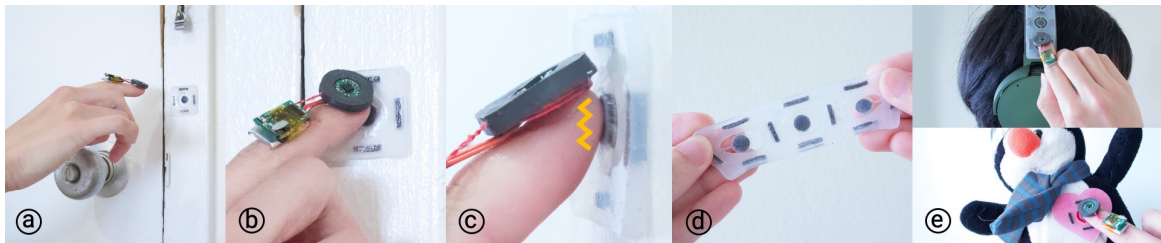


Figure 1: (a) We demonstrate a new type of haptic actuator, which we call MagnetIO, that is comprised of two parts: *any number* of soft interactive patches that can be applied anywhere and *one* battery-powered voice-coil worn on the user's fingernail. (b) When the fingernail-worn device contacts *any* of the interactive patches it detects its magnetic signature and (c) makes the patch *vibrate*. (d) To allow these otherwise passive patches to vibrate, we make them from silicone with regions doped with neodymium powder, resulting in soft and stretchable magnets. (e) This novel decoupling of traditional vibration motors allows users to add interactive patches to their surroundings by attaching them to walls, objects or even other devices or appliances without instrumenting the object with electronics.

ABSTRACT

We demonstrate a new type of haptic actuator, which we call MagnetIO, that is comprised of two parts: one battery-powered voice-coil worn on the user's fingernail and any number of interactive soft patches that can be attached onto any surface (everyday objects, user's body, appliances, etc.). When the user's finger wearing our voice-coil contacts any of the interactive patches it detects its magnetic signature via magnetometer and vibrates the patch, adding haptic feedback to otherwise input-only interactions. To allow these passive patches to vibrate, we make them from silicone with regions doped with polarized neodymium powder, resulting in soft and stretchable magnets. This stretchable form-factor allows them to be wrapped to the user's body or everyday objects of various shapes. We demonstrate how these add haptic output to many situations, such as adding haptic buttons to the walls of one's home.

KEYWORDS

soft magnets, ubiquitous haptics, fabrication

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1 INTRODUCTION

Today's interactive devices increasingly instrument every kind of surface, effectively adding interactive functionality even to passive everyday objects such as walls, tables, and rapidly prototyped objects [1, 4, 12–14], as well as to the user's body [6, 7, 11]. To enable sensing these interactions, researchers engineered conformable/stretchable sensing devices so that these can comfortably fit around non-planar surfaces, which is the case for everyday objects or the human body. This led to a mature field of on-body or on-object *sensing* technologies that are easy to deploy and ideal for prototyping or enabling interactions in ubiquitous settings.

However, the converse is not the case, while sensing can be done in a distributed fashion around the user and in conformable form-factor, the same is not true for *actuation*. Researchers are still looking for techniques that allow deploying large numbers of actuators without the constraints of power delivery to every single individual actuator, wireless communication across all actuators, microcontrollers, etc. As a result, while we have a range

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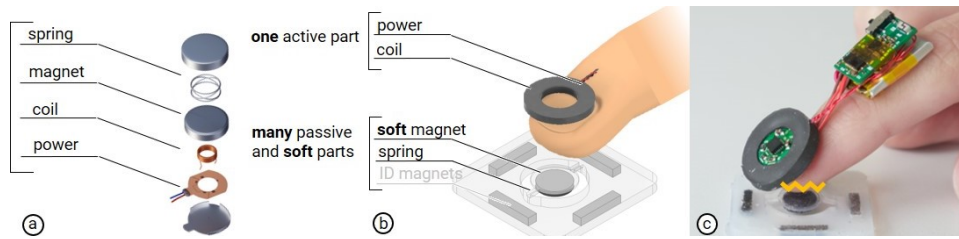


Figure 2: (a) Our device is built from the same principle as traditional linear resonant actuators (LRA), i.e., based on moving a magnetic mass using a coil and a spring. (b-c) However, our device decouples these components into two parts: *one* active element containing the coil, which the user wears on their fingernail; and, *many* passive elements, which are comprised of silicone with regions doped with neodymium powder which together realize a spring & magnet.

of interactive techniques to deploy sensing “patch”-like devices everywhere, these patches typically do not exhibit any form of haptic response, i.e., they can sense the user’s touch (e.g., to turn on/off the user’s home-alarm) but they cannot vibrate in response to that touch (e.g., to indicate the alarm is on/off), not at least, without requiring vibration motors or other haptic actuators, which in turn require batteries and circuitry. Ultimately, all these dramatically limit the ubiquitous application of these interactive patch-like devices.

In this demonstration, we display a new alternative for adding haptic feedback to everyday surfaces, which is depicted in Figure 1. Our approach, which we call *MagnetIO*, introduces a new type of haptic actuator that is passive (i.e., requires no electronics, no battery, etc.) until the user’s finger, which is instrumented with a wearable voice-coil, touches it, causing it to vibrate.

2 OUR APPROACH: MAGNETIO

MagnetIO is composed of *many* passive interactive patches and *one* nail-worn device, which features a miniaturized and custom-engineered voice-coil, inertial measurement unit (IMU), battery, microcontroller and wireless. MagnetIO’s complete voice-coil and circuitry fits entirely on the user’s fingernail, thus leaving the fingerpad unobstructed to feel interactions with the user’s environment and the vibrations from our passive patches.

The design principle that enables our interactive patches to vibrate is that they are made from silicone doped with neodymium powder, resulting in stretchable magnets that are attracted and repelled by the wearable coil. Conceptually, we liken our approach to a linear resonant actuator (LRA, typical vibration element in most commercial mobile devices), which is comprised of a magnet attached on a spring and a coil, which we depict in Figure 2

To better illustrate our design, we make an analogy to the inner workings of an LRA: when current is supplied to the LRA’s coil, it produces a magnetic field which attracts or repels the magnet. By performing this multiple times, the momentum of this *magnet-spring* assembly creates a feelable vibration (Figure 2a).

In our case, our design takes a sharp conceptual turn from that of the LRA as we *purposely decouple* the magnet-spring from the coil (Figure 2b). This allows us to scale up the output by placing *many* spring-magnet pairs everywhere, which are activated when the user’s wearable coil contacts them (Figure 2c). The advantage is that our design requires only *one* coil, meaning it also only requires

one driver circuit, *one* communication module and *one* battery. Furthermore, all our interactive patches are passive since we implement them using silicone, an elastic material that allows us to achieve the “spring” component; subsequently, some regions are doped with neodymium powder, which allows us to realize the “magnet.” Therefore, our soft-magnets realize the “spring-magnet” component typical of LRAs. The result is that MagnetIO patches are easy to place anywhere because they are stretchable and soft, and yet they deliver haptic output to the user, appearing as interactive patches. Details on the implementation and characteristics of our device can be found in [5].

3 WALKTHROUGH: ADDING MAGNETIO PATCHES EVERYWHERE

To give the reader a complete picture of how MagnetIO allows a user to control their environment with ad-hoc interactive haptic patches, we demonstrate a simple application using our MagnetIO patches to control an internet-of-things (IoT) device at the user’s home.

Figure 3a shows our user, wearing our wearable voice-coil on their index finger, walking into their home. Figure 3b shows that as they walk in, they tap a MagnetIO patch that has been attached to their door. In response to tapping the patch, they feel two consecutive vibrations (*tzzz*, pause, *tzzz*) confirming that their home-alarm is now disabled, which is depicted in Figure 3c. Note that this user can also perform this action *in the dark* (i.e., eyes-free, which we did not illustrate for the sake of visual clarity) because our MagnetIO patches are inherently haptic interfaces designed to vibrate on touch.



Figure 3: (a) Our user at home, wearing our coil, surrounded by surfaces with interactive patches. (b) They tap their wall, which has a passive patch that controls their home alarm. (c) The user feels the patch vibrate to indicate that their alarm is now disabled.

Now that we depicted an interaction with MagnetIO from the user's perspective, let us examine what is happening from the device's perspective. In other words, we will describe how our patches work to deactivate the user's IoT home alarm. Figure 4 depicts the principle behind MagnetIO interactions. First, as depicted in Figure 4a, as the user taps on the patch to deactivate their home-alarm, the wearable-coil approaches the patch and senses its ID, i.e., the wearable-coil recognizes that the particular patch the user is touching is the "home-alarm" ON/OFF patch. To sense the ID of a patch, the wearable-coil makes use of its inertial measurement unit, which includes a 3DOF magnetometer. Using the magnetometer data, the wearable-coil detects the patch that the user is interacting with by comparing the current reading to pre-trained magnetic signatures. These magnetic signatures are just an example of one many possible ways our system could sense the ID of each patch (refer to *Implementation* for details). Alternatively, our system could utilize a wide variety of sensing mechanisms, such as RFIDs [2, 8], acoustic IDs [3], optical IDs [9, 10], etc. After the wearable-coil has identified that the user touched the "home-alarm" patch, it communicates to the user's IoT home-alarm via its Bluetooth module, informing it to switch to the "OFF" state. Finally, as depicted in Figure 4b, the home-alarm confirms the new state by sending a message to the wearable-coil, which the wearable device renders *haptically* by energizing the coil in a vibrotactile pattern. This creates a magnetic field which in turn attracts the magnetically-doped region of the interactive patch, making the patch vibrate under the user's finger.

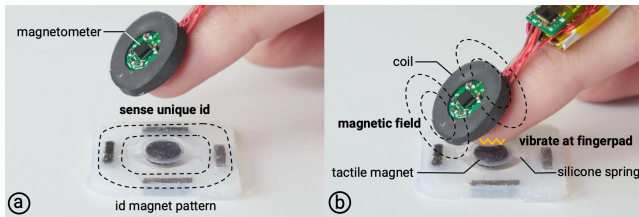


Figure 4: (a) When the user's finger approaches an interactive patch, its magnetometer reads and recognizes its unique 3D magnetic ID and (b) activates the wearable-coil, generating a magnetic field that vibrates the patch under the fingerpad.

Next, as depicted in Figure 5a, the user taps on another patch on the wall to adjust their thermostat. Figure 5b, depicts how the user

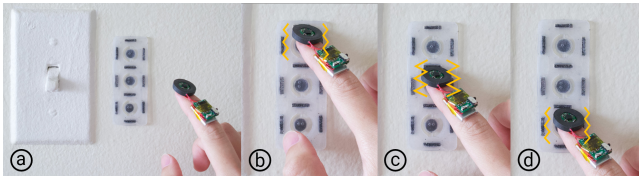


Figure 5: The user touches the top magnet and slides their finger down to the middle magnet where they feel the strongest vibration, indicating the last setting used and continue to slide down to the bottom to reset the thermostat to the lowest setting.

slides their finger across the patches, at each step, they feel a short vibration that haptically signals each temperature level. However, as depicted in Figure 5c, when they cross the middle patch, they feel a strong vibration (a haptic detent) that indicates that this is the last used setting.

MagnetIO patches are versatile because they are *passive*, made from stretchable silicone. Figure 6a illustrates how our user detaches a patch from the wall and attaches it to new objects. These patches are sticky as their underside is made of a layer of skin safe adhesive, making our patches also suitable for ad-hoc on-body interfaces. In fact, MagnetIO patches are even weather-proof. For instance, Figure 6d depicts a user washing their smart kettle instrumented with an interactive patch. Because MagnetIO's patches operate via magnetism alone (for both sensing and actuation, no circuitry is inside the patch), the interaction will not be affected so long as the wearable coil remains dry.



Figure 6: (a) The user peels off the patch. Because our interactive patches are *passive* and *soft*, they can be applied to a variety of objects: (b) smartwatch bracelet, (c) water bottle, (d) and even washed (still works while wet).

Besides our sticker patch, we also implemented a strap-like patch, depicted in Figure 7, that can be wrapped around objects, such as handles, bottles, etc. These types of patches are ideal for cylindrical objects such as the gaming controller depicted in Figure 7a. In Figure 7, our user stretches the patch around the grip of the controller, which enhances its functionality by allowing them to mode-switch in their game, while keeping their eyes on the game menu; again, an eyes-free interaction.



Figure 7: (a) The user holds a strap-like patch and (b) stretches it around their gaming controller, enabling (c) an eyes-free I/O space for controlling modes and menus.

Now, the user's partner, who is Blind, walks into the room. Despite being visually-impaired, their partner also makes use of MagnetIO patches to control their home. Since MagnetIO are interactive *haptic* patches, they can serve as useful interfaces for visually-impaired users. Especially, because MagnetIO patches are simply cast from silicone, one can add *tactile bumps* that encode messages in Braille, as depicted in Figure 8. This allows our visually-impaired user to use the thermostat interface by adding a side patch with Braille annotations. The user then feels the Braille to know which

patch this is e.g., “thermostat” or “setting 1”, and so forth (Figure 8b), and then make use of MagnetIO’s vibrations to know they have selected a setting on the interface (Figure 8c).

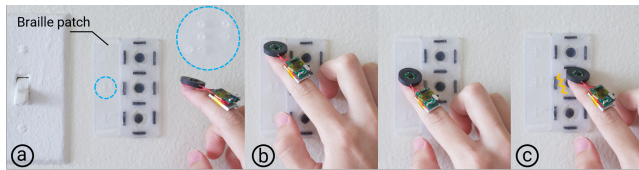


Figure 8: Since our interactive patches are made from silicone, they may easily include Braille to assist Blind users.

While our walkthrough exhibits the key principles behind how MagnetIO allows to deploy haptic patches everywhere, these do not depict an exhaustive list. We designed MagnetIO to be attached to many more objects and believe that it will inspire researchers to use, or even create, passive haptics patches that can be anywhere.

4 CONCLUSION

We demonstrated MagnetIO, a new type of haptic actuator comprised of two parts: *one* battery-powered voice-coil worn on the user’s fingernail and *many* of interactive soft patches that can be attached onto any surface (everyday objects, user’s body, appliances, etc.). When the user’s finger wearing our coil contacts any of the interactive patches it detects its magnetic signature via magnetometer and makes the patch vibrate, adding haptic feedback to otherwise input-only interactions. To allow these passive patches to vibrate, we fabricated them from silicone with regions doped with polarized neodymium powder, resulting in soft and stretchable magnets. This stretchable form-factor allows our patches to be wrapped to the user’s body or everyday objects of various shapes.

It is this novel technical implementation, based on the decomposing the inner-workings of a linear resonant actuator, that gives rise to MagnetIO’s unique feature, it is a *one-to-many haptic device*, i.e., *one* active part (an electromagnetic-coil worn the fingernail) powers *many* passive actuators.

We demonstrated a range of applications that make use of MagnetIO’s patches to realize ubiquitous haptics, i.e., surfaces and objects around the user can now exhibit interactive behavior (they can vibrate upon touch).

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